

TABLE OF CONTENTS

M-002 PIPE SUPPORTS AND HANGERS

	Page No.
1 SCOPE OF WORK	M 002-2
2 CODES AND STANDARDS	M 002-2
3 GENERAL REQUIREMENTS.....	M 002-3
3.1 The analysis shall consider	M 002-4
3.2 Insulation Weights.....	M 002-5
4 DETAILED REQUIREMENTS	M 002-6
4.1 General	M 002-6
4.2 Spring Type Units.....	M 002-9
4.2.1 Travel Stops and Locks.....	M 002-10
4.3 Piping Attachments	M 002-11
4.4 Auxiliary Steel	M 002-11
4.5 Materials.....	M 002-11
4.6 Loading	M 002-12
4.7 Other Detailed Requirements	M 002-13
4.8 Anchors, Guides And Restraints	M 002-14
4.9 Snubbers.....	M 002-14
4.10 Supports For Small Piping And Tubing	M 002-17
4.11 Tools	M 002-17
4.12 Preparation For Delivery	M 002-17
4.13 Nameplates	M 002-18
5 CLEANING AND PAINTING	M 002-18
6 QUALITY ASSURANCE / QUALITY CONTROL	M 002-18

M 002 PIPE SUPPORTS AND HANGERS

1 SCOPE OF WORK

The Bidder shall supply the following in accordance with Part 4, Subsection 1.2 Summary of Work.

All hangers, anchors, auxiliary steel, gussets, supports, guides, clamps, rods, clevises, turnbuckles, braces, brackets and ties, complete with all components as required for all the piping systems furnished by the Bidder.

All necessary auxiliary steel for pipe supports to connect to the plant structures shall be furnished by the Bidder. The terms “piping supports”, “piping hangers”, or “pipe supports” as used herein shall comprise the entire range of various methods and devices used to carry and support the piping systems, their valves, other piping components, fluid transported, insulation, appurtenances and seismic and transverse loading.

2 CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5, ‘Codes and Standards’. The codes and standards of the following organizations extracted from Part 4, Subsection 2.5 “Codes and Standards” or other international equivalent standards are specifically applicable to the design, manufacture and testing of the equipment included in this specification.

AISC	American Institute of Steel Construction
ANSI	American National Standards Institute
ASTM	American Society of Testing and Materials
ASME	American Society of Mechanical Engineers
AWS	American Welding Society
MSS	Manufacturers Standardization Society of the Valve and Fitting Industry
DIN	Deutsches Institut für Normung

Supports for all piping systems shall be design, fabricated, and installed in accordance with ANSI B31 .1 latest edition including addenda unless otherwise noted.

3 GENERAL REQUIREMENTS

Pipework shall be supported and anchored in an appropriate manner in accordance with the provisions of ASME Standard ANSI B31.1 or the Standard to which the pipework is designed.

Supports shall not be dependent for flexibility on the flexure of supporting rods or straps. Supports shall be positioned before the pipe has been erected and near to valves wherever possible. Supporting straps around flanges or welded joints will not be accepted.

Pipe on racks and in the trenches should not rest directly on structures. They shall have independent supports. Teflon or similar sheet should be used between two stub plane.

All supports shall be provided with means of adjusting the tension of springs, the height of supports and the length of rods or straps. All such adjusting devices shall be capable of being securely locked.

Parts of supporting elements which are subjected principally to bending or tension loads or working temperatures for which carbon steel is not recommended, shall be made of suitable alloy steel or shall be protected so that the temperature of the supporting member will be maintained within the appropriate temperature limits of the material.

The supporting arrangements of all pipework systems shall be designed with due regard to any additional loads imposed during hydraulic or hydrostatic leak testing, and cleaning and steam blowing operations.

Particular attention shall be paid to sliding surfaces in guides and sliding supports and to the springs employed in variable and constant load supports. The Bidder shall provide stainless steel and low friction plastic for important sliding surfaces.

Where necessary, constant load supports shall be provided.

The Bidder shall take all necessary precautions to ensure that the piping shall be free from vibration by the installation of sway braces and/or vibration dampers or other acceptable means.

Except for springs, all supports shall be designed for a safety factor of not less than 5, based on the ultimate strength of the material.

Outdoor piping support systems shall be designed to withstand the environmental conditions described elsewhere in this specification. Spring supports should be avoided in such situations otherwise they shall be weatherproofed, the springs being plastic coated or otherwise protected.

Supports shall be so designed and fabricated that, in the event of spring failure, they do not release the pipe. Springs shall be of the helical compression type and shall be provided with means to prevent misalignment, buckling, eccentric loading or excessive travel, with a working stress not exceeding 10 N/mm² at the most favorable loading conditions.

The Bidder shall prepare, and include in the plant operating and maintenance manuals, pipework layout or other similar drawings marked to show the position of every pipe support on all high pressure pipework, and any pipe $\geq$ DN 80.

Each of the supports marked on these drawings shall be assigned a unique plant reference number by the Bidder. The Bidder shall prepare, for submission to the Engineer before erection, and for inclusion in the operating and maintenance manuals, a schedule of all these supports. The support schedule shall, as a minimum, state the support reference number, the type of support, the directions of freedom of movement, and the magnitude and direction of pipe movement from the cold installed to the hot operating conditions. A spring support schedule shall also be provided to the Engineer and included in the operating manuals, which as a minimum, states the support reference numbers, the spring rates, the hot operating load, and the magnitude and direction of travel from the cold installed to the hot operating conditions.

The Bidder shall verify by calculation that the piping systems meet the requirements of ANSI B31 .1 with respect to thermal expansion and flexibility.

A comprehensive stress analysis shall be performed. For all steam and hot water system the following:

3.1 The analysis shall consider

- a Pipe (and equipment) cold with full account being taken of cold pull incorporated in the line and due allowance being made for the dead

weight of the piping and insulation. Tabulate maximum cold terminal reactions.

- b Pipe hot with two thirds of actual cold pull incorporated in the line and due allowance being made for the dead weight of the piping, fluid and insulation. Tabulate maximum hot terminal reactions.
- c Pipe hot with no allowance for cold pull and with due allowance for dead weight. For this calculation the modulus of elasticity will be “cold” (i.e. as erected) value for the material being used. Tabulate position and magnitude of maximum expansion stress and compare with the allowable stress range as defined in ANSI B31.1 latest edition.

The following information shall also be tabulated from the above calculations:

- a The displacement of pipe elements from the “as erected” position to the post-cold pull position and from the post-cold pull position to the final hot position.
- b The hot, cold and hydro-test loads of the system to be supported at each hanger position.

Flexibility and hanger load calculations shall be based on actual pipe dimensions.

Flexibility analysis shall be made for each mode of operation of the equipment and the piping shall be designed for the worst case.

Piping systems connected to items of plant shall be designed to meet terminal loadings on such items of plant to within plant manufacturer’s acceptable figures under all conditions of operation.

If exposed to direct sunlight, large diameter piping for natural gas, fuel oil or other cold fluids shall be analyzed by the Bidder to protect the piping systems against thermal expansion during shut down.

3.2 Insulation Weights

Insulation weight, in the design of hangers, supports, anchors etc. shall be calculated based on the insulation thickness and lagging required .

Piping insulation shall be calcium silicate having a density of approximately 224 kg/m³.

Note : An additional weight shall be allowed for aluminum lagging applied on all

insulated piping of 1.12 kg/m² for 0.4 mm thick aluminum.

Bidder shall submit detailed pipe support drawings to the Owner for approval. Pipe support drawings shall be submitted after the stress analysis, where applicable.

Pipe support drawings shall reference the specific sheet of the stress analysis which shows the pipe loads and movements, where applicable.

Pipe support drawings shall show the following:

- (a) All design loads and moments imposed by the piping at the support. Cold, hot, hydro test and seismic loads shall be shown, as applicable.
- (b) All pipe thermal movements shall be shown.
- (c) All shop and field welds shall be shown.
- (d) All supplementary steel for the support and reinforcement of plant structural steel members shall be shown.

Forces and moments imposed by piping at equipment and terminal points shall be tabulated by the Bidder and submitted to the Owner for information. The tabulated list shall reference drawings which show the allowable force and moments as determined by the equipment supplier or supplier responsible for the terminal point.

The location and magnitude of loads imposed on structural steel due to pipe supports shall be included in the analysis for plant structural steel as the design basis for supplementary steel referenced above.

4 DETAILED REQUIREMENTS

4.1 General

The supports shall be designed to meet all static dynamic, and operational conditions to which the piping system and connected equipment will be subjected. These conditions shall include hydrostatic test loadings, thermal

expansion and contraction, seismic loading impact, vibration, and (for outdoor locations) wind loadings. For supports specified to be designed in accordance with ANSI B31 .1 the loads shall not be based on increasing the allowable stress level in hanger components by 20% as allowed by Paragraph ASME B31.1 121 .2.G.1 of the Code. Bidder may use engineering model to determine the design for all Bidder's piping supports.

Pipe support for non-standard wall piping shall not be released for fabrication until the actual dimensions and weights of such piping have been determined and shown on the piping list and design drawings.

The type and thickness of insulation will be per Part 4, M-003 Technical Specification, Thermal Insulation and Lagging. The weight of the insulation shall be indicated on the design drawings and shall include weight of jacket or covering. Release of support for fabrication shall not be done before the thickness and weight of insulation are confirmed and indicated.

No pipe shall be supported from another pipe. However, multiple runs of cold pipe may be supported together on trapeze hangers. Pipes shall not be hung from cable trays or cable tray hangers. Pipe support rods shall not pass through cable trays.

Continuously threaded support rods are not acceptable when rod diameter is greater than 9 mm. Welding of extension pieces to support rods is prohibited for all size rods.

All guide type supports shall be designed so that piping cannot become disengaged by normal pipe movements.

Bidder shall provide adequate clearance for all systems in both hot and cold positions between supports and any obstructions. For system where thermal growth occurs, sufficient clearance shall be provided to prevent interference between supports and surrounding equipment during this growth.

Bidder shall not support pipe off grating, horizontal or vertical bracing, trolley beams or minor steel. Minor steel shall be defined as flange width (b_f) less than:

- 12.5 mm for wide flange and M shapes
- 62.5 mm for S shapes

- 55 mm for C shapes
- 75 mm for MC shapes

Building steel supporting more than one hanger shall have all hangers attached to the beam's flange. Stiffeners shall be provided, where required, on all plant structural member where pipe support are attached.

Unless otherwise indicated no expansion anchor or through bolts shall be installed at the following locations:

- The bottom of concrete slabs on metal deck.
- Masonry and hollow concrete block walls.
- All (hot, cold, and hydro) support loads shall be shown on the design drawings and shall include the weight of support assemblies.

Owner reserves the right to require Bidder to modify the pipe supporting elements with respect to type, location, loading, movement and quantities prior to manufacture.

Bidder shall provide the services of a hanger service engineer at Site just prior to and during initial operation of each unit to inspect and readjust the installation of all pipe supporting element units furnished by Bidder for proper installation.

All piping hangers for piping in sizes 80 mm and larger shall be designed so that they are adjustable (changes in piping elevation made) without the necessity of moving all or part of the load from the hanger.

Bearing surfaces and bolt holes shall not be flame cut.

All horizontal movement of piping 25 mm or greater shall be shown on each detail hanger drawings.

Design of pipe supporting elements located outdoors shall include wind loads affect of up to 165 kg/m² per ANSI wind load criteria. Units for insulated piping located outdoors shall be provided with galvanized hanger rod flashing and rain hoods. The galvanization thickness shall be as specified in specification M 001 Piping.

All components shall be designed with a minimum factor of safety of 5 based on the ultimate strength of the material.

All devices and turnbuckles shall be forged steel and compatible with the material of rod. Turnbuckles and lockouts shall be provided, if not supplied internally, with constant support and variable spring hangers. Hanger rods fastened to supporting steel members or embedded steel using clevis with lockout attached to inserts or to plate, angle iron or tie iron, or using beam clamps, shall be subject to review and acceptance by the Owner.

Welding to steel members shall comply with the detail drawings. Use of U-type members welded on each side of a beam web will not be permitted. Where steel for hangers or supports is to be welded to embedded steel plates, the steel shall be located and restricted so that the outside edge of the weld is a minimum of 25 mm from the edge of the embedded steel plate, in each case.

The supporting element for rigid vertical support units shall be designed so that supporting rods are within 0.5 degree from the vertical, in the hot position. Pipe clamps or beam connections shall permit the horizontal movement of pipe (when hot) to swing the hanger rod into the vertical position.

Saddles shall be provided for any bottom-supported (trapeze) horizontal lines having an insulation thickness over 25 mm. Rod attachments at clamps shall be clear of the insulation.

4.2 Spring Type Units

Bidder shall furnish all necessary variable spring and constant support spring hangers to accommodate all vertical thermal movements for steam and hot water piping systems.

Spring hangers shall be designed for installation in the cold position such that the hanger rod is vertical when pipe moves to the hot operating position. The maximum vertical offset between pin-to-pin connections when cold shall be 4 degrees.

Variable-support units shall have enclosed springs and an indicator showing the actual load and movement. Constant-support and variable-support units shall have an indicator showing hot and cold positions.

Where load variation exceeds the limitations specified below, constant support hangers must be used. For these applications variable spring hangers shall not be used:

- a. When load variation between hot load and cold load exceeds 6 % of the cold load for main steam, and high pressure auxiliary steam and turbine bypass system.
- b. When load variation between hot load and cold load exceeds 10 % of the cold load for all piping systems having design temperature higher than 200° C
- c. When load variation between hot and cold load exceeds 25 % for all piping system having design temperature less than 200° C.

Constant-support-type units shall have a minimum adjustable supporting weight margin of plus or minus 10% from the calculated load, and an over travel margin of not less than 25% or 25 mm whichever is greater.

4.2.1 Travel Stops and Locks

All constant-support and variable-support units shall be provided with hydrostatic test and erection locks, installed and preset in the cold position on the units before they are shipped. The locks must be removable and replaceable in a cold position and shall be provided with permanent means of attaching them to the support unit when not in use. These devices shall be painted red and have an attached large “CAUTION” tag stating that the device must be removed before putting the pipe line into service.

All constant and variable support units shall be designed and constructed with the following provisions;

- a. Over travel in either direction shall be stopped before the spring(s) would be overstressed.
- b. Failure of spring(s) shall not cause the unit to release the load.
- c. All compression coil springs shall have ends ground flat and shall be provided with suitable guides where necessary to prevent bowing when loaded.
- d. Spring-support construction shall be such that the hanger rod or spring compressing rod will be maintained concentric with the spring.

Where spring-type units are to be installed outdoors, the units shall be designed to protect the springs and working parts from corrosion and rain.

Where required, rigid hanger supports shall be provided in sufficient quantities to support such piping properly. Rigid hanger supports shall be of ample strength, made of steel and of a type.

4.3 Piping Attachments

The piping attachments shall be of the same material as the piping to which they are attached by welding. All clips, lugs, plates, saddles etc. Shall be welded to piping.

Attachments shall be welded in accordance with the requirements of the governing piping code by qualified welders.

All straps, ring hangers, clamps etc. fastened to piping by means other than welding may be of carbon steel material, where design temperature of piping is greater than 260°C.

Austenitic stainless on alloy steel U-bolts shall be used in the supports for piping where design temperature is greater than 260°C and shall be solution-annealed after hot or cold forming.

4.4 Auxiliary Steel

All supplementary steel required for connecting pipe supports to structural steel, or concrete shall be furnished by the Bidder.

4.5 Materials

Plate material shall be ASTM A36, or approved equal.

Steel pipe when used or stand support shall be ASTM A106, Grade B or ASTM A53, Grade B. If not welded to the piping system.

All bolted connections shall be made with ASTM A325 bolts in a friction type connection.

All slotted connections to accommodate thermal movement shall be made with A307 shoulder bolts or A325 bolts snug-tight with double nuts.

All welded connections shall be made with E70XX electrodes or approved equal.

All sliding bearing plates shall be lubricated or Teflon plates with a maximum coefficient of friction equal to 0.10.

4.6 Loading

All loads resulting from stress analysis and seismic conditions shall be considered as acting concurrently and their effects shall be considered in design of pipe support-and auxiliary steel.

Hydro, Test Loads - Associated with hydrostatic testing shall be considered in the design of pipe support components and auxiliary steel.

4.7 Other Detailed Requirements

Auxiliary support steel consisting of double channels shall be used for vertically loaded supports. Auxiliary support steel subject to biaxial bending, torsion, axial forces or any combination of these due to applied piping loads shall consist of wide-flange sections or structural tube sections.

Single angle and single channel shall not be used for supports having load of 200 kg and more.

Where it is necessary to weld to existing structural steel members to carry pipe supports, anchors, etc., the welds for securing the plates to the structural members shall in all cases be parallel to the length and located at web of the structural member.

Welding across the flange of existing structural steel members is not permitted.

If pipe supports are cantilevered from building structural beams or channels, due design consideration shall be taken to avoid unacceptable torsion on building structural steel beams or channels.

All supplementary steel provided by Bidder shall be attached to existing framing by means of connection angles to ensure proper fit up and avoid rigid web to web connections.

The Bidder shall not attach supplementary support steel to cover plates on existing steel framing.

If a knee-braced support or a cantilevered support is attached to structural steel column, Bus plate shall be provided.

Bidder shall furnish and install a pair of angle stiffeners on existing structural steel members when the attachment of supplementary support steel induces local flange bending in the existing support steel.

Bidder's responsibility to verify the actual locations of the building Steel prior to fabrication of auxiliary steel. Any modifications required shall be at Bidder's expense.

4.8 Anchors, Guides And Restraints

Anchors, guides and restraints shall be designed for the imposed loading for anchor design shall be submitted to the Owner for review and approval.

The Bidder shall provide earthquake loading for anchors, guides and restraints for all piping.

Hydraulic snubbers or similar devices shall be provided in the system to allow sufficient flexibility to accommodate seismic loading.

Piping subject to shock loadings, such as those generated by quick closing valves, water hammer and relief or safety valve reaction forces, shall be provided with shock absorbing devices.

All braces when required to eliminate piping vibration shall be furnished by Bidder.

Insulated piping shall be provided with standard piping covering protection saddles except where special designs are furnished.

Anchors, guides and restraints shall be designed so that heat will not be transferred from the piping to the building steel. The temperature of the support member connected to the building steel shall be less than 55°C.

4.9 Snubbers

Snubbers shall be provided to restrain and dampen the imposed loading when subjected to abnormal rapid motion without locking in a fixed position or without interfering with normal, slowly occurring thermal expansion movement.

Snubbers shall permit continued pipe movement in either axial direction at a constant velocity of not less than 25 mm per minute, while restraining a sustained applied load. The snubber shall move freely when subjected to axial motion at velocities less than 75 mm per minute. The resistance to movement shall not exceed 5 kg for units rated up to 500 kg load or 1 % of rated load for units with rated loads greater than 500 kg.

Each snubber assembly shall contain spherical ball bushings that permit a minimum total of 100 angular rotation of the snubber in any direction.

The total free axial movement, excluding that due to fluid compressibility and structural parts shall not exceed 0.8 mm for dynamic cyclic loading between 2 and 10 cycles per second. The 0.8 mm includes mechanical gaps, valve movements and other movements that are not in a linear relationship with size of applied load.

Test data shall be submitted that will demonstrate that the snubbers can resist the specified loads without undergoing excessive displacements. The test data shall include a full description of the tests conducted and evidence that the requirements of this Specification are met.

Hydraulic snubbers shall be used wherever practical in preference over mechanical snubbers. The hydraulic snubber shall meet following specific requirements.

- (a) Hydraulic fluid shall be nonflammable. Fluid shall be suitable for the environmental requirements indicated in this specification.
- (b) During operation the snubber piston shall be free to move unrestricted in either direction (i.e., tension or compression), with the popped valves remaining fully open for all piston velocities up to 250 mm/mm. This velocity is greater than any operational thermal growth and less than the velocity of any normally anticipated disturbing force. Both the compression and tension controlling popped valves shall be designed and set for closure when the piston velocity exceeds 250 mm/mm.
- (c) With the popped valves closed the fluid flow shall stop, thereby transforming the unit into a rigid strut. Closure is designed to be effected within a piston rod movement of 0.8 mm. After the disturbing forces subside, the popped valves shall open to again allow free thermal movement of the piping. The design shall preclude the occurrence of a ratcheting effect which would cause stepwise buildup of excessive pipe displacements. Evidence which proves that ratcheting will not occur shall be provided. Play shall be minimized through the use of close-tolerance fittings.
- (d) The reservoir system shall be designed to permit installation of the snubber in any orientation. Over a two-year period, the total leakage of the in-service fluid shall not exceed 10% of the fluid volume in the snubbers, exclusive of the reservoir system. The reservoir shall be sized to accommodate twice the design leakage rate. The use of transparent

reservoirs in the primary equipment cells will not be acceptable. Design shall permit monitoring the level of the reservoir fluid.

- (e) Snubbers shall have a hard chromed, stainless steel piston rod working in conjunction with a bronze rod bushing.
- (f) The assembly shall have ball-joint attachment and shall be designed so that it is capable of exerting the required force in compression and tension, utilizing the distance between the pivot of the pipe attachment and the pivot of the steel attachments as the effective assembly length without the use of the telescoping guides. Minimum available piston stroke shall be equal to the anticipated pipe movement plus 50 percent in the installed condition with a minimum stroke of 125 mm.
- (g) All parts of the snubbers and component shall be so fabricated that they will not become disengaged by thermal movement of the supported pipe.
- (h) All springs furnished for outdoors locations shall be coated with neoprene. All other ferrous metal components (including alloy) furnished that will be-located out-of-doors, except bearing surfaces and surfaces to be field welded, shall be galvanized or of other corrosion resisting materials subject to review and acceptance by the Owner. Surfaces to be field welded shall not be galvanized. Galvanized surfaces will not require painting.

Mechanical-type snubbers shall have all stainless steel working components, including internals and bearing and shall not require the use of lubricants. Housing, brackets and struts may be of carbon steel. All working parts shall be enclosed within a protective metal housing. Mechanical snubbers shall restrain applied loads and arrest motion when retracted or extended at an acceleration of 0.02g to a maximum of 0.1g.

4.10 Supports For Small Piping And Tubing

All small piping such as instrument sensing, sampling and control piping and tubing, capillary leads etc. which are subject to sag or damage by construction personal, plant operators or maintenance personal shall be laid on channels or shall have continuous supporting means such as Tube Track, Unistrut or other suspension system.

All suspension hardware shall be of materials and assembly design which provide f or protection against galvanic corrosion.

For outdoor service the supporting hardware shall be galvanized with galvanic corrosion protection provisions.

4.11 Tools

Bidder shall furnish a complete set of all special tools required to adjust and dismantle all supports. Tools shall be new. Tools shall be shipped to the job in a separate container clearly marked with the name of the equipment for which the tools are intended. These tools shall not be used for erection purposes and - shall be delivered to the Owner's warehouse.

4.12 Preparation For Delivery

All shop-fabricated supports shall be packaged and securely packed in containers for shipment in a manner that will ensure arrival at destination in satisfactory condition.

All assembled units shall be identified at the shop by means of a metal tag or other permanent material securely attached to the unit with wire. Where supports are not shipped as assembled units, each component or subassembly shall be properly tagged with the item number or other identifying information. Tags shall remain on the units after erection, as a permanent record. All random materials shall be tagged for identification. All parts from each support shall be packaged in a separate container. Each container shall be marked with the, item number and other identifying information. If more than one support is shipped in a single container, each item shall be securely packed to prevent damage during transit. The contents of container shall be itemized and labeled as previously described.

All moving parts requiring lubrication shall have the lubricant applied in the shop.

4.13 Nameplates

Each spring support and snubber unit shall have a metal nameplate attached in a conspicuous place. The following information shall be clearly stamped thereon:

- a. Manufacturer's Model and/or Catalog Number.
- b. Manufacturer's Serial Number.
- c. Size.
- d. Load (in kg). as follows:
 1. Calibrated load for snubbers.
 2. Cold and hot loads for variable supports.
 3. Operating load for constant supports.
- e. Owner's or Bidder's Identification Number. (KKS Numbers)
- f. Travel (in mm), where applicable.
- g. "Caution" or "Danger" information.
- h. Lubrication instructions.

Nameplates shall be protected to avoid damage during shipment, storage and erection.

5 CLEANING AND PAINTING

All material shall be free of loose scale, rust, oil, grease and other deleterious foreign materials. All unmachined ferrous surfaces shall be painted or coated in accordance with the requirement as mentioned in M-001, S-420 and S-430.

6 QUALITY ASSURANCE / QUALITY CONTROL

Quality Assurance and Quality Control shall conform to the requirements as mentioned in the Specification of Part 4 Subsection 2.7.

(Last page of M 002)